

ZAHRA ABBASIANAE

 zahra-abbasiantaeb  ZahraAbbasiantaeb  z.abbasiantaeb@uva.nl
 Lab 42, Science Park, Amsterdam  Website  Google Scholar

RESEARCH INTERESTS

I am a Machine Learning Scientist and Ph.D. Candidate at the University of Amsterdam (graduating April 2027), specializing in Natural Language Processing (NLP) and Information Retrieval (IR). My research covers key aspects of agentic conversational search systems, including retrieval, LLM-based evaluation, personalization, synthetic data generation, and user simulation. I have published over 10 papers in these areas, alongside a proven track record of building production-scale NLP pipelines for Booking.com and the European Patent Office (EPO) at Leiden University. Additionally, as a co-organizer of the TREC iKAT track, I have helped publish reusable benchmarks, evaluation methodologies, and tools for information-seeking systems.

EDUCATION

University of Amsterdam (UvA) <i>Ph.D. Candidate in Artificial Intelligence</i> Focus: Agentic Conversational Search and Personalization	Amsterdam, Netherlands Expected April 2027
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Amirkabir University of Technology <i>M.Sc. in Artificial Intelligence and Robotics - Summa cum laude</i> Focus: Question Answering Systems and Deep Attentional Neural Networks	Tehran, Iran 2018 – 2021
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Amirkabir University of Technology <i>B.Sc. in Software Engineering - Summa cum laude</i>	Tehran, Iran 2014 – 2018
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WORK EXPERIENCE

Booking.com <i>Machine Learning Scientist Intern</i>	Amsterdam, The Netherlands 2025
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- Designed and deployed a domain-agnostic synthetic data generation pipeline capable of bootstrapping Machine Learning (ML) models in zero-annotation settings, eliminating reliance on human-labeled data.
- Developed novel style-diversity techniques to increase synthetic dataset variance and model generalization, leading to a paper currently under review [1].
- Built scalable generation workflows using PyTorch, Hugging Face (TRL/PEFT) to support dataset generation across multiple enterprise use cases.

Leiden University (LIACS) <i>Researcher (In Collaboration with European Patent Office (EPO))</i>	Leiden, Netherlands 2022
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- Designed an end-to-end reference extraction pipeline, including reference detection, field detection, and entity matching to link in-text references in patents to Web of Science (WoS) scientific publications.
- Curated human-annotated datasets to fine-tune and evaluate NLP models such as transformer-based language models for sequence labeling for different stages of the designed pipeline.
- Deployed the pipeline across complete patent corpora from both the European Patent Office (EPO) and USPTO, generating a large-scale mapping dataset connecting global patents to scientific literature.
- Published the full architecture and dataset in *Information Processing & Management* (2025), establishing an enterprise-grade citation-matching standard [10].

SKILLS

Core Expertise	Agentic Conversational Search & IR, Large Language Models (LLMs), RAG, Alignment & Preference Optimization, Synthetic Data Generation, Evaluation
Programming	Python, Java, C/C++, R, SQL
ML & LLM Frameworks	PyTorch, Hugging Face (Transformers, TRL, PEFT, Datasets), DeepSpeed, Bitsandbytes, vLLM, TensorFlow, Scikit-learn, Numba
Data & Vector Engineering	Elasticsearch, PostgreSQL, MySQL, MongoDB, Pandas, NumPy
MLOps & Infrastructure	Docker, Kubernetes, Git, AWS, SLURM/HPC, Linux (SysAdmin)
API & Web Deployment	FastAPI, Flask, REST APIs

BOOKS

Momtazi, Saeedeh, and **Zahra Abbasiantaeb** Question Answering over Text and Knowledge Base. Springer Nature, 2022. [\[pdf\]](#)

PUBLICATIONS

1. **Zahra Abbasiantaeb**, Zeno Belligoli, Omar Essam, and Mohammad Aliannejadi. The Significance of Style Diversity in Annotation-Free Synthetic Data Generation. Under review. [\[pdf\]](#)
2. **Zahra Abbasiantaeb***, Simon Lupart*, Marcel Gohsen, Nailia Mirzakhmedova, Johannes Kiesel, Jeffrey Dalton, and Mohammad Aliannejadi. TREC IKAT 2025: A Test Collection for the Offline and Interactive Evaluation of Conversational Search. *SIGIR*, 2026. [\[pdf\]](#)
3. Mahta Rafiee, Heydar Soudani, **Zahra Abbasiantaeb**, Mohammad Aliannejadi, Faegheh Hasibi, and Hamed Zamani. Total Recall QA: A Verifiable Evaluation Suite for Deep Research Agents. *SIGIR*, 2026. [\[pdf\]](#)
4. Marcel Gohsen, Nailia Mirzakhmedova, **Zahra Abbasiantaeb**, Johannes Kiesel, Simon Lupart, Jeffrey Dalton, Benno Stein, and Mohammad Aliannejadi. Sim. API: A Middleware to Simplify the Use of User Simulators for Shared Tasks in Conversational Search. Under review, 2026. [\[pdf\]](#)
5. **Zahra Abbasiantaeb**, Simon Lupart, and Mohammad Aliannejadi. Generating Multi-Aspect Queries for Conversational Search. *EACL*, 2026. [\[pdf\]](#)
6. **Zahra Abbasiantaeb***, Simon Lupart*, Leif Azzopardi, Jeffrey Dalton, and Mohammad Aliannejadi. Conversational Gold: Evaluating Personalized Conversational Search System using Gold Nuggets. *SIGIR*, 2025. [\[pdf\]](#)
7. Maryam Mousavian, **Zahra Abbasiantaeb**, Mohammad Aliannejadi, and Fabio Crestani. Towards Fair Rankings: Leveraging LLMs for Gender Bias Detection and Measurement. *ICTIR*, 2025. [\[pdf\]](#)
8. Clemencia Siro*, **Zahra Abbasiantaeb***, Yifei Yuan, Maarten de Rijke, and Mohammad Aliannejadi. Do Images Clarify? A Study on the Effect of Images on Clarifying Questions in Conversational Search. *CHIIR*, 2025. [\[pdf\]](#)
9. **Zahra Abbasiantaeb**, Chuan Meng, Leif Azzopardi, and Mohammad Aliannejadi. Improving the Reusability of Conversational Search Test Collections. *ECIR*, 2025. [\[pdf\]](#)
10. **Zahra Abbasiantaeb**, Suzan Verberne, and Jian Wang. Tracing science-technology-linkages: A machine learning pipeline for extracting and matching patent in-text references to scientific publications. *Information Processing & Management* 62.6, 2025: 104264. [\[pdf\]](#)
11. Mohammad Aliannejadi, **Zahra Abbasiantaeb**, Shubham Chatterjee, Jeffrey Dalton, and Leif Azzopardi. TREC IKAT 2023: A Test Collection for Evaluating Conversational and Interactive Knowledge Assistants. *SIGIR*, pp. 819-829, 2024. [\[pdf\]](#)
12. **Zahra Abbasiantaeb**, Yifei Yuan, Evangelos Kanoulas, and Mohammad Aliannejadi. Let the llms talk: Simulating human-to-human conversational qa via zero-shot llm-to-llm interactions. *WSDM*, pp. 8-17, 2024. [\[pdf\]](#)

13. Emiliana Murgia, **Zahra Abbasiantaeb**, Mohammad Aliannejadi, Theo Huibers, Monica Landoni, and Maria Soledad Pera. ChatGPT in the Classroom: A Preliminary Exploration on the Feasibility of Adapting ChatGPT to Support Children’s Information Discovery. *UMAP*, pp. 22-27, 2023. [\[pdf\]](#)
14. Kasra Darvishi, Newsha Shahbodaghkhan, **Zahra Abbasiantaeb**, and Saeedeh Momtazi. PQAD: A Persian question answering dataset. *Computer Speech & Language* 80, 2023: 101486. [\[pdf\]](#)
15. **Zahra Abbasiantaeb**, Suzan Verberne, and Jian Wang. Optimizing BERT-based reference mining from patents. *PatentSem Tech*, 2022. [\[pdf\]](#)
16. **Zahra Abbasiantaeb** and Saeedeh Momtazi. Entity-aware answer sentence selection for question answering with transformer-based language models. *Journal of Intelligent Information Systems* 59.3, 2022: 755-777. [\[pdf\]](#)
17. **Zahra Abbasiantaeb** and Saeedeh Momtazi. Text based question answering from information retrieval and deep neural network perspectives: A survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 11.6, 2021: e1412. [\[pdf\]](#)

HONORS AND AWARDS

Ranked 1st in TREC iKAT 2024 Competition	2024
Nominated for Apple Scholars in AI/ML PhD Fellowship	2024
Received ACM SIGIR Student Travel Grant	2024
Ranked 1st in TREC iKAT 2023 Competition	2023
Received WIR (Women in IR) SIGIR Registration Grant	2022
Ranked 1st in M.Sc. cohort based on GPA, Amirkabir University of Technology, Iran	2020
Invited Scholar for Exchange Program, University of Passau, Germany	2020
Ranked 5th in B.Sc. cohort based on GPA, Amirkabir University of Technology, Iran	2018
Offered Direct Admission to M.Sc. program for academic excellence, Amirkabir University	2017
Ranked 212th among 70,000+ students in the National University Entrance Exam, Iran	2014

ACADEMIC SERVICES

Co-organizer: TREC Interactive Knowledge Assistant Track (iKAT).	2023–2026
Conference Reviewer: CLEF, SIGIR, CIKM, ECIR.	2024-2026

TUTORIALS

Title: Query Understanding in LLM-based Conversational Information Seeking	July 2025
Conference: SIGIR 2025, Padua, Italy.	

INVITED TALKS

LLMs in Conversational Search: From Query Understanding to Evaluation , University of Glasgow.	December 2025
User Simulation in Conversational Search , Qazvin Islamic Azad University (QIAU), Iran.	May 2024

TEACHING EXPERIENCE

Information Retrieval 1, M.Sc. , University of Amsterdam (UVA)	Summer 2026
Information Retrieval 0, B.Sc. , University of Amsterdam (UVA)	Fall 2024
Information Retrieval 0, B.Sc. , University of Amsterdam (UVA)	Fall 2023
An introduction to Natural Language Processing (NLP) , Hamrah Academy	Summer 2021
Information Retrieval and Web Search, M.Sc. , Amirkabir University of Technology (AUT)	Fall 2019
Machine Learning, M.Sc. , Amirkabir University of Technology (AUT)	Spring 2019
Artificial Intelligence, B.Sc. , Amirkabir University of Technology (AUT)	Spring 2018
System Design and Analysis, B.Sc. , Amirkabir University of Technology (AUT)	Fall 2016
Advanced Computer Programming, B.Sc. , Amirkabir University of Technology (AUT)	Spring 2016
Principals of Computer and Programming, B.Sc. , Amirkabir University of Technology (AUT)	Fall 2015